

VVU HBK Mk-II — Phase 2 Power & Thermal Architecture Engineering Specification

8S4P 32700 LiFePO₄ Battery • Star Ground Wiring • Epistemic Thermal Governance

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1. System Overview

The HBK Mk-II is a ruggedized, Bayesian inference-capable edge terminal designed for municipal water infrastructure monitoring. The system operates within a 460×360 mm working volume, mounted on a 3.0 mm CNC-machined 6061-T6 aluminum base plate within an IP67-rated enclosure. The architecture integrates high-performance compute, precision analog sensing, and robust power management in a thermally constrained, electrically isolated package.

Key design principles include epistemic thermal governance (four-tier temperature response), star-ground power distribution (P0–P3 rail isolation), and mandatory EMI/RFI exclusion zones between analog and power domains. The 8S4P 32700 LiFePO₄ battery pack provides 614 Wh of field-sustainable energy with a Daly 8S 20A BMS for cell-level protection.

2. CAD Module Layout

All modules are positioned relative to the base plate origin (0,0,0) at the lower-left corner. Z-coordinates reference the top surface of the 3.0 mm base plate.

Module	Material / Type	Dimensions (mm)	Position (x,y,z)
Base Plate	6061-T6 Aluminum	460 × 360 × 3	(0, 0, 0)
AMD Ryzen AI Compute Engine	Compute Module	140 × 130 × 45	(160, 120, 3)
Analog Front-End / Sensor Shield	PCB + Shield	120 × 160 × 22	(20, 180, 3)
PM-01 Power Dist. & Daly 8S 20A BMS	Power Module	110 × 140 × 38	(20, 20, 3)
8S4P 32700 LiFePO ₄ Pack	Battery Pack	150 × 85 × 80	(140, 20, 3)
Pyrogel XTE Aerogel Thermal Isolation	Thermal Barrier	160 × 95 × 85	(135, 15, 3)
NVMe Storage Bay	Storage Module	40 × 90 × 15	(320, 60, 3)
Sealed Comms Routing Node	Comms Module	100 × 140 × 25	(340, 200, 3)

3. 8S4P 32700 LiFePO₄ Battery Specification

The HBK Mk-II power source is an 8S4P pack constructed from IFR-32700 cylindrical LiFePO₄ cells. LiFePO₄ chemistry provides superior thermal stability, flat discharge voltage, and long cycle life (≥2000 cycles at 80% DoD) — critical for unattended municipal deployments.

Parameter	Value
Cell Type	IFR-32700 cylindrical, 3.2 V nominal
Configuration	8S4P (32 cells total)
Pack Voltage (Nominal)	25.6 V (8 × 3.2 V)
Pack Voltage (Range)	20.0–29.2 V
Pack Capacity	20 Ah (4 × 5 Ah parallel)
Pack Energy	614 Wh (25.6 V × 20 Ah × 1.2 ruggedization)
BMS	Daly 8S 20A — cell-level monitoring, balancing, protection
Max Discharge	20 A continuous (1C)
Ruggedization Overhead	15% — busbars, cell holders, epoxy potting
Operating Temperature	–10°C to +60°C (charge limited 0–45°C)

4. Star Ground Wiring Protocol (P0–P3)

The HBK Mk-II implements a four-rail star-ground architecture to prevent ground loops, minimize conducted EMI, and ensure analog signal integrity. Each rail originates from a single star point on the PM-01 power distribution module.

Rail	Designation	Wire Gauge	Path / Description	Isolation
P0	Main Power Rail	10 AWG	Battery → BMS, high-current DC	Direct
P1	System Power Rail	14 AWG	Compute power distribution	Filtered
P2	Clean Rail	18 AWG	Galvanically isolated via DC-DC converter	Galvanic
P3	Signal Rail	STP*	Physically separated from P0/P1	Physical + Shield

* STP = Shielded Twisted Pair. P3 signal cabling maintains ≥50 mm physical separation from P0/P1 power conductors throughout the enclosure.

5. Epistemic Thermal Governance

The HBK Mk-II implements a four-tier epistemic thermal governance framework. Each tier defines a mandatory system response triggered by measured temperature thresholds. The governance model is “epistemic” in that it reasons about system state under uncertainty — sensor readings are treated as evidence, not ground truth, and the system conservatively escalates when thermal evidence exceeds advisory thresholds.

Tier	Temperature	Classification	System Response
T0	22°C	Ambient	Baseline operating temperature — normal operation
T1	65°C	Advisory	Increased monitoring, reduced compute load, log escalation
T2	75°C	Warning	Graceful service degradation, active cooling engagement
T3	85°C	Critical	Emergency shutdown, data preservation, battery disconnect

6. Thermal Containment Architecture

Thermal containment is achieved through a four-layer architecture that directs heat from the compute engine outward to the enclosure while isolating the battery pack and analog front-end from thermal contamination.

Layer	Mechanism	Thermal Conductivity	Function
1	TIM PCM	5–7 W/m·K	Phase-change thermal interface — absorbs transient heat spikes
2	Structural Conduction	—	Mainboard-to-enclosure thermal path via base plate
3	Aerogel Isolation (Pyrogel XTE)	0.015 W/m·K	Ultra-low-conductivity barrier around battery pack
4	Enclosure Radiation	—	IP67 shell as final thermal boundary — radiative + convective

The Pyrogel XTE aerogel layer (0.015 W/m·K) provides approximately 400× thermal resistance compared to the TIM PCM layer, ensuring that battery pack temperature remains within safe operating limits even during sustained compute workloads.

7. Bill of Materials (BOM)

#	Component	Specification	Qty	Source	Category	Status
1	IFR-32700 LiFePO ₄ Cell	3.2 V, 5 Ah, cylindrical	32	EVE / Lishen	Battery	Qualification
2	Daly 8S 20A BMS	8S, 20 A continuous, UART	1	Daly Electronics	BMS	In Production
3	6061-T6 Base Plate	460×360×3 mm, CNC machined	1	Custom Fab	Structural	Design Review
4	AMD Ryzen AI Compute Module	Ryzen AI + 16 GB LPDDR5	1	AMD	Compute	Evaluation
5	Analog Front-End PCB	8-ch 24-bit ADC, anti-aliasing	1	Custom PCB	Sensing	Prototype
6	Pyrogel XTE Insulation	160×95×85 mm, 0.015 W/m·K	1	Aspen Aerogels	Thermal	In Stock
7	TIM PCM Pad	5–7 W/m·K, phase-change	4	Honeywell / Laird	Thermal	In Stock
8	IP67 Enclosure	500×400×150 mm, alum alloy	1	Hammond / Custom	Enclosure	Design Review
9	NVMe SSD (Industrial)	512 GB, -40–85°C rated	1	Samsung / WD	Storage	In Stock
10	DC-DC Isolated Converter	24→12 V, 30 W, galvanic isolation	2	Murata / Traco	Power	In Stock
11	Sealed Comms Module	RS-485 + LTE + LoRa, IP67	1	Custom / Sierra	Comms	Prototype
12	Wiring Harness Kit	10/14/18 AWG + STP, pre-terminated	1	Custom Assembly	Electrical	Design Review

8. EMI/RFI Isolation

The 15.0 mm physical isolation gap between the Analog Sensor Interface PCB and the Power Management Unit is a mandatory design constraint. This exclusion zone is enforced to prevent conducted and radiated EMI from the power domain from corrupting sensitive analog measurements.

The following constraints apply within the 15.0 mm EMI/RFI exclusion zone:

- No conductive traces of any kind
- No via fencing or via stitching
- No copper pour (ground or power planes)
- No ferromagnetic components
- Physical separation enforced by base plate routed channel

VIOLATION: Any conductive element within the 15.0 mm exclusion zone constitutes a critical design failure and requires immediate rework. The EMI isolation gap is verified by automated DRC (Design Rule Check) in the CAD toolchain and by physical inspection during assembly.